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CLOTHING WITH BLANK FIELDS TO AFFIX CUSTOM ARTWORK USING INDIRECT AND DIRECTION SUBLIMATION PRINTING

Abstract

An article of clothing and a method of making the same. The article of clothing includes a white or light-colored area or panel or field suitable for sublimation printing, surrounding at least in part by an area of fixed, permanent background color or pattern. An image is heat transferred to the article of clothing in registration with the printable field such that there is no discernible boundary between the printed image and the permanent background color or pattern. The heat-transferable image may be a image disposed on a heat transfer sheet that can be transferred to an article of clothing by the application of heat and pressure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] Priority is hereby claimed to provisional application Ser. No. 63/561,865, filed Mar. 6, 2024, which is incorporated herein by reference.

BACKGROUND

[0002] Custom clothing in some form or another has been a staple of commerce for thousands of years. Prior to the industrial revolution, all clothing was, in a sense, custom made. The modern era has given rise to large-scale machinery for making thread, weaving or knitting the thread into fabric, cutting the fabric into appropriately sized pieces, and sewing the pieces into a final article of clothing. And yet there remains a robust market for short-run customized clothing—for example sports team uniforms (everything from Little League to Major League Baseball and every league in between). The uniforms are typically customized with each player's name and number. Similarly, the market for customized or decorated industrial uniforms is robust. Another example is “wearable art”; that is, customized clothing bearing copyrighted images of specific artists.

[0003] Current methods to meet those needs, though, are not ideal. Embroidered, heat-pressed and sewn-on patchwork give clothing a very stylish and high-quality customized appearance. However, embroidery requires expensive computer-controlled embroidery equipment and a skilled workforce to operate that machinery. Sewn-on patchwork requires that the needed patches be maintained in inventory. It also requires a skilled workforce to sew or heat press the patches in place.

Additionally, heat-pressed patches have a tendency to peel from the surface of clothing over time and after many washings.

[0004] Screen printing, also known as silk-screening, is a printing technique in which a mesh is used to transfer ink (or dye) onto a substrate except in areas made impermeable to the ink by a blocking stencil. A blade or squeegee is moved across the mesh or “screen” to fill the open mesh apertures with ink. A reverse stroke then causes the screen to touch a substrate (such as an article of clothing). This causes the ink to wet the substrate and be pulled out of the mesh apertures as the screen springs back after the blade has passed. Only one color can be printed per screen and per pass of the blade. Therefore several screens, applied one after another, in registration with each other, and using a different color of ink for each screen, must be used to produce a multi-colored image.

[0005] Thus, silk-screening is quite limited in terms of the artwork that can be applied. Single-pass silk screening can only apply a monochromatic work of art, such as a line drawing or printed names or other alpha-numeric indicia.

[0006] What the market needs is a fast, easy method to customize a shirt or other article of clothing that does not require expensive equipment, or highly skilled operators, and which does not requiring maintaining a physical inventory of custom patches or silk screens. Additionally, the method should yield an article of clothing that does not appear to have been assembled in a slap-dash or assembly-line fashion. But rather, the final product should appear to have been custom made from beginning to end.

SUMMARY OF THE INVENTION

Disclosed and claimed herein are the following:

[0007] A method of affixing indicia or artwork to an article of clothing. The method comprises providing an article of clothing comprising at least one, permanently defined or attached printable field having a shape, a size, and a perimeter, wherein the printable field is lighter in color than a surrounding area of the article of clothing, and wherein the surrounding area of the article of

clothing is a fixed, pre-fabricated color or pattern.

[0008] Providing a heat-transferable image having a shape, a size, and a perimeter, wherein: the shape of the heat-transferable image corresponds to the shape of the printable field; the size of the heat-transferable image is the same as or larger than the size of the printable field; and the perimeter of the heat-transferable image is a color or pattern substantially identical to the fixed, pre-fabricated color or pattern of the surrounding area of the article of clothing.

[0009] Heat transferring the heat-transferable image in registration onto the printable field to yield a heat-transferred image having a perimeter, wherein there is no visually discernible border between the perimeter of the heat-transferred image and the perimeter of the printable field.

[0010] Another version is a method of affixing indicia or artwork to an article of clothing, the method comprising providing an article of clothing comprising at least one, permanently defined or attached printable field having a shape, a size, and a perimeter, wherein the printable field is lighter in color than a surrounding area of the article of clothing, and wherein the surrounding area of the article of clothing is a fixed, pre-fabricated color or pattern, and wherein the perimeter of the printed field comprises a color or grey-scale gradient.

[0011] Providing a heat-transferable image having a shape, a size, and a perimeter, wherein the shape of the heat-transferable image corresponds to the shape of the printable field; the size of the heat-transferable image is the same as or larger than the size of the printable field; and the perimeter of the heat-transferable image comprises a color or grey-scale gradient inverse to and in registration with the color or grey-scale gradient on the perimeter of the printable field of the article of clothing.

[0012] Heat transferring the heat-transferable image in registration onto the printable field to yield a heat-transferred image having a perimeter, wherein there is no visually discernible border between the perimeter of the heat-transferred image and the perimeter of the printable field.

[0013] Yet another version of the method comprises providing an article of clothing comprising at least one, permanently defined or attached printable field having a shape, a size, and a perimeter, wherein the printable field is lighter in color than a surrounding area of the article of clothing, and wherein the surrounding area of the article of clothing is a fixed, pre-fabricated color or pattern, and wherein the perimeter of the printed field comprises a decorative motif.

[0014] Providing a heat-transferable image having a shape, a size, and a perimeter, wherein: the shape of the heat-transferable image corresponds to the shape of the printable field; the size of the heat-transferable image is the same as or larger than the size of the printable field; and the perimeter of the heat-transferable image comprises a decorative motif corresponding to and in registration with the decorative motif on the perimeter of the printable field of the article of clothing.

[0015] Heat transferring the heat-transferable image in registration onto the printable field to yield a heat-transferred image having a perimeter, wherein the decorative motif on the perimeter of the heat-transferable image is transferred in registration onto the decorative motif on the perimeter of the printed field.

[0016] In any of the methods disclosed herein the heat-transferable image may be provided on a heat transfer sheet.

[0017] An article of clothing produced by any of the above methods.

[0018] Also disclosed and claimed herein is an article of clothing comprising at least one, permanently defined or attached printable field having a shape, a size, and a perimeter, wherein the printable field is lighter in color than a surrounding area of the article of clothing, and wherein the surrounding area of the article of clothing is a fixed, pre-fabricated color or pattern.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1A depicts an exemplary shirt (here a short-sleeved, collared, button-up shirt) from the front (left panel) and from the back (right panel) wherein the front panels and sleeves are fabricated from pre-dyed fabric in conventional fashion, but the back panel includes an undyed (or white-dyed) panel, area, or printable field **16** that is dimensioned and configured to be a substrate for digital printing using dye sublimation inks. The printable field **16** has a boundary or perimeter **16'** that is sharply defined.

[0020] FIG. 1B depicts the back of the exemplary shirt of FIG. 1A in conjunction with an image printed on a transfer sheet **19**. As shown in the figure, the image on the transfer sheet **19** is transferred to the printable field **16**. The final product, shown in the lower-right panel, has no perceptible border between the printed image **22** and the pre-existing background **18**.

[0021] FIG. 2A depicts another exemplary shirt from the front (left panel) and from the back (right panel). The back panel includes an undyed (or white-dyed) panel, area, or printable field **16** that is dimensioned and configured to be a substrate for digital printing using dye sublimation inks. The printable field **16** has a boundary or perimeter **24** that comprises a color gradient from the undyed or white color of the field **16** to the pre-existing background color **18**.

[0022] FIG. 2B depicts the back of the exemplary shirt of FIG. 2A in conjunction with an image printed on a transfer sheet **19**. As shown in the figure, the image on the transfer sheet **19** is transferred to the printable field **16**, such that the edges of the image on the transfer sheet **19** are in registration with the color gradient **24**. The final product, shown in the lower-right panel, has no perceptible border between the printed image **22** and the pre-existing background **18**.

[0023] FIG. 3A depicts yet another exemplary shirt from the front (left panel) and from the back (right panel). The back panel includes an undyed (or white-dyed) panel, area, or printable field **16** that is dimensioned and configured to be a substrate for digital printing using dye sublimation inks. The printable field **16** has a boundary or perimeter **30** that comprises a decorative frame **30**.

[0024] FIG. 3B depicts the back of the exemplary shirt of FIG. 3A in conjunction with an image printed on a transfer sheet **19**. As shown in the figure, the image on the transfer sheet **19** is transferred to the printable field **16**, such that the edges of the image on the transfer sheet **19** are in registration with the decorative frame **30**.

[0025] FIG. 4A depicts a hat **40** that includes an undyed (or white-dyed) panel, area, or printable field **16** that is dimensioned and configured to be a substrate for digital printing using dye sublimation inks.

[0026] FIG. 4B depicts the hat of FIG. 4A in conjunction with an image printed on a transfer sheet **19**. As shown in the figure, the image on the transfer sheet **19** is transferred to the printable field **16**, such that the edges of the image on the transfer sheet **19** are in registration with the edges of the printable field **16**.

[0027] FIG. 4C depicts the final product—the hat **40** with the image **42** printed on it such that there is no perceptible border or transition area between the image **42** and the pre-existing background **18**.

DETAILED DESCRIPTION

[0028] Numerical ranges used herein are intended to include every number and subset of numbers contained within that range, whether specifically disclosed or not. Further, these numerical ranges should be construed as providing support for a claim directed to any number or subset of numbers in that range. For example, a disclosure of from 1 to 10 should be construed as supporting a range of from 2 to 8, from 3 to 7, from 1 to 9, from 3.6 to 4.6, from 3.5 to 9.9, and so forth.

[0029] All references to singular characteristics or limitations of the present method shall include the corresponding plural characteristic or limitation, and vice-versa, unless otherwise specified or clearly implied to the contrary by the context in which the reference is made. The words “a” and “an” mean “one or more” unless specifically defined to the contrary. The word “or” is used

inclusively and means “and/or.”

[0030] All combinations of method or process steps disclosed herein can be performed in any order, unless otherwise specified or clearly implied to the contrary by the context in which the referenced combination is made.

[0031] The methods disclosed herein can comprise, consist essentially of, or consist of the essential elements or limitations or steps of the method described herein, as well as any additional or optional steps, ingredients, components, or limitations described herein or otherwise useful in sublimation printing onto a fabric substrate.

[0032] Throughout all of the drawings, the same reference numerals are used to designate the same or similar elements.

[0033] Heat-transferring an image to an existing article of clothing is conventional. Typically, a darker image is heat-transferred onto a lighter-colored article of clothing. This is done to prevent the color of the garment itself from skewing the colors of the heat-transferred image. While lighter images can be heat-transferred onto a darker-colored article clothing, the results are often not acceptable to the buying public. This is because the colors in the image being transferred are altered by the darker color of the cloth substrate to which it is being applied. This makes the ultimate product either amateur in appearance or lacking in crisp colors or contrast.

[0034] In the present method (and in the product produced by the method), a garment such as the shirt **10** in FIG. **1A** or the hat **40** in FIG. **4A** is manufactured in conventional fashion using pre-dyed colored fabric. However, at least one portion or panel **16** of the garment is a field dimensioned and configured for sublimation printing. The printable field **16** is surrounded by a background **18** of any pre-determined color or pattern. The printable field **16** is either uncolored fabric or fabric that has been colored to be white, nearly white, or generally lighter in shade than the background **18**. The printable field **16** is suitable for sublimation printing.

[0035] Referring now specifically to FIGS. **1A** and **1B**, shown in FIG. **1A** is an article of clothing, in this instance an exemplary shirt **10**. The front **12** of the shirt **10** is depicted in the left-hand panel and the back **14** of the same shirt **10** is shown in the right-hand panel of FIG. **1A**. As shown in the left-hand panel, the front **12** of the shirt **10** may be of conventional construction, with two breast pockets, plackets with buttons and corresponding buttonholes, and a collar (all unnumbered). (See below, though, for further discussion.) The right-hand panel, which shows the back **14** of the shirt **10**, has printed on it, defined within it, attached to it, or disposed or positioned on it, at least one printable field **16**. (Optionally, as shown in the left-hand panel of FIG. **1A**, the pockets, for example, could also optionally comprise a printable field **16**, or the sleeves, or the collar, or any other portion of the article of clothing **10**.) The printable field **16** is positioned within a pre-existing and fixed background fabric **18** of a pre-determined, pre-fabricated, and arbitrary color or pattern. The printable field **16** is a permanently incorporated into or attached to the article of clothing **10**; it is not a detachable element. The permanent attachment can be accomplished by any means now known or developed in the future, such as (but not limited to) sewing, rivets, fusion, permanent adhesives, and the like. As shown in the right-hand panel of FIG. **1A**, the border **16'** between the printable field **16** and the fixed background **18** is discrete, sharp, and well-defined. As shown in the right-hand panel of FIG. **1A**, there is no gradient or fade between the printable field **16** and the background **18**. The object then of this first version of the method is to print, via sublimation printing, a desired image within the printable field **16** such that there is no visible discernable border or distinction between the field **16** and the fixed background **18** after the desired image has been printed within the printable field **16**.

[0036] In the present method, the fabric of the printable field **16** is either left undyed or is a light- or white-colored panel. An undyed or white printable field **16** is preferred because it yields the most vivid colors via sublimation printing. The printable field **16** can be sewn into the article of clothing at the time of its original manufacture or otherwise permanently defined, disposed, or positioned within or adjacent to the fixed background **18** of the article of clothing. As noted

previously, the printable field **16** is permanently incorporated into or otherwise permanently attached to the article of clothing. That is, the shirt shown in FIGS. **1A** and **1B** includes at least one permanent panel, area, portion, etc. that comprises a printable field **16**. The printable field **16** is dimensioned and configured to be a substrate for digital printing using sublimation inks or dyes or any other type of sublimation colorant.

[0037] Of note is that article of clothing **10** is manufactured to include at least one printable field **16** from the original manufacturer of the shirt. This pre-made garment, with an unprinted printable field **16** as shown in FIG. **1A**, is part of the present disclosure. The unprinted article of clothing **10**, including at least one printable field **16** can be sold, as is, either wholesale to clothing customizers or retail to end users who would then heat-transfer their own artwork to the white-colored panel. This allows the buyers of the shirt to apply their own artwork to the printable field **16** via conventional heat transfer techniques. Additionally, because the unprinted shirts **10** as shown in the right-hand panel of FIG. **1A** can be manufactured in bulk, but customized and sold profitably in very short runs, the method disclosed herein reduces the risk of accumulating unsaleable inventory. The image to be printed into printable field **16** is completely customizable and at the entire whim of the buyer.

[0038] The at least one printable field **16** may appear anywhere on the article clothing. Thus, for example, in FIGS. **1A**, **2A**, and **3A**, a printable field appears on both the front and rear of the garment. This is simply a design choice. The article of clothing could have only one printable field **16** located anywhere on or in the garment. Thus, the article of clothing disclosed herein may comprise, consist essentially of, or consist of only one printable field **16**, or the article of clothing may comprise a plurality of printable fields **16**. All such articles of clothing are explicitly encompassed by the present disclosure. This, for example, in FIG. **1A**, the pockets in the left-hand panel may comprise (in whole or in part) a printable field **16**.

[0039] The preferred method for printing the image onto the printable field **16** is sublimation printing. Sublimation printing comprises several distinct digital computer printing techniques that use heat to transfer dye onto a substrate (nominally via sublimation of a solid dye, as the name implies). As used herein, the term “sublimation printing” applies to all direct-to-substrate and indirect sublimation printing methods (now known or developed in the future). Thus, for example, one sublimation transfer printing process uses a conventional ink-jet printer to deposit ink onto a transfer sheet. The printed transfer sheet is then pressed against the fabric to be printed and heat is applied to the transfer sheet, the substrate, or both. The heat functions to transfer the dye from the transfer sheet to the fabric (nominally via sublimation). Thus, this implementation of sublimation printing is deemed “indirect” because the fabric substrate itself does not pass through a printer.

[0040] The term “direct dye sublimation” is sometimes applied to a variant of digital textile printing using dye-sublimation inks printed directly onto a fabric substrate, which is then heated to set the dyes. The adjective “direct” indicates that a transfer sheet is not used. As dye sublimation page printers utilize heat to transfer the dye onto the print media, the printing speed is limited by the speed at which the elements on the thermal head can change temperature. Heating the elements is easy, as a strong electric current can raise the temperature of an element very quickly. However, cooling the elements when changing from a darker to a lighter color, is more technically demanding. It usually involves having a fan or a heat sink assembly associated with the print head to cool it more quickly. Machines that use multiple print heads are also known. This speeds up the printing process as one or more print heads cool, while one or more other print heads are actively printing directly to the substrate.

[0041] Sublimation transfer printing is a digital printing technology using full-color artwork that works best with polyester fabrics and polymer-coated substrates.

[0042] For best results when printing onto a fabric substrate via dye sublimation, the fabric used should be comprised mostly or (or entirely of) polyester. Other, non-polyester substrates can be printed, but the image produced will not be as vivid. In the conventional process, the desired image

is first printed onto coated, heat-resistant transfer paper as a mirror-image of the final design. The image is then transferred from the transfer sheet onto the substrate using a heat press. Early large-format printers used for printing onto transfer paper were modified electrostatic plotters that used toner as the ink/dye. More recently, large-format sublimation printers are essentially inkjet printers modified to use special inks. For small-format printing, inkjet has also become the dominant technology, though special dye-sublimation transfer laser printers are also commercially available. There are a host of commercial manufacturers of dye-sublimation printers capable of printing onto a fabric substrate, including (among many others) HP Inc. (until 2015, the Hewlett Packard Company, Palo Alto, California, USA) and Seiko Epson Corporation (Suwa, Nagano, Japan).

[0043] To transfer an image from a sheet or roll of transfer paper to a fabric substrate, a heat press machine, typically programmable for time, temperature, and pressure, is used. Different settings are used depending on the fabric. Via a suitable combination of time, temperature, and pressure, the heat press transfers the sublimation dyes from the transfer sheet to the substrate. The most common conventional dyes currently used for sublimation printing activate at a temperature of about 350° F. (175° C.). However, a temperature range of from about 380° F. to about 420° F. (about 195° C. to about 215° C. is normally recommended to achieve optimal color of the transferred image on the fabric substrate.

[0044] The result of the sublimation transfer process is a highly durable, colorfast, high-resolution, full-color image on the substrate. Because the dyes are infused into the substrate (essentially at the molecular level), rather than applied at a topical level (such as with screen printing and direct-to-garment contact printing), the printed images does not crack, fade, or peel from the substrate (under normal conditions). The fabric is permanently dyed so it can be washed without damaging the quality of the image. The advantages of dye-sublimation printing over other methods of textile printing are thus clear. The images are permanent and do not peel or fade. The dye does not build up on the surface of the fabric. Colors can be extraordinarily brilliant due to the bonding of the dye to transparent fibers of certain synthetic fabrics. Truly continuous tones can be achieved that are equivalent in resolution to a photograph (without the need for special techniques, such as half-screen printing). The image can be printed anywhere on the substrate, including all the way to the edges.

[0045] Referring now to FIG. 1B specifically, this figure demonstrates the method disclosed herein using indirect sublimation printing onto a transfer sheet **19**, and then transferring the image from the transfer sheet to an article clothing. (Here, for illustration only, the article of clothing is the shirt **10** of FIG. 1A, and the printable field **16** is on the back of the shirt against a darker, fixed background **18**.) The top panel of FIG. 1B shows a transfer sheet **19** having a back side **20** and a front side or printed side) **20'**. The image to be transferred to the printable field **16** is printed, mirror-image style, on the front side **20'** of the transfer sheet **19**. As can be seen from the upper panel of FIG. 1B, the image printed on the transfer sheet includes a peripheral color or pattern **18'** that matches the fixed background color **18** of the garment to be printed.

[0046] The lower-left panel of FIG. 1B shows the transfer sheet positioned in registration with (and optionally slightly overlapping) the printable field **16**, with the printed side **20'** of the transfer sheet in face-to-face contact with the printable field **16**. Heat and pressure are then applied to the back side **20** of the transfer sheet to transfer the image printed on the front side **20'** of the transfer sheet to the printable field **16**. Because the peripheral color or pattern **18'** on the transfer sheet is closely matched to the fixed background color or pattern **18** of the garment itself, the final product, shown in the lower-right panel of FIG. 1B has no visually discernible border between the printed image **22** and the fixed background color or pattern **18**. To the casual (or not so casual) viewer, it appears that the image **22** was fabricated into the garment **10** from the outset as a monolithic panel (rather than printed into the printable field **16**).

[0047] In this fashion, garments that include one or more printable fields **16** can be very quickly customized, in either small runs or large runs. The garments with one or more blank printable fields

16 can be sold in bulk (to reduce inventory costs), but then custom printed on an “as-needed” and “just-in-time” basis. This allows for the customizer to fulfill a custom order at great speed and at greatly reduced cost.

[0048] This is especially helpful in the uniform business, where articles of clothing conventionally incorporate a host of fixed elements juxtaposed with custom elements. Athletic team uniforms are a perfect example. The team name is a fixed element that appears on all of the uniform shirts. That team name could be, for example, a fixed patch sewn onto every shirt, or an image permanently embroidered onto every shirt. But each uniform shirt is then customized with at least a number unique to each player. Some teams, such as many National Football League teams, also include each player's name across the shoulder panel of the shirt. This is a troublesome proposition for custom clothing manufacturers because each player is a different size. So for the uniform maker, each player's unique number and unique name must be fabricated into an article of clothing of the appropriate size for that individual player. This creates inventory bottle necks and inefficiencies because the makers want to fill orders quickly (to satisfy their customers), but at the same time do not want to maintain bloated inventories of semi-modified pieces of clothing of every conceivable size and color combination (i.e., uniforms pieces with the team name permanently affixed, but no player number or player name). The problem is compound, for example, when teams have different “home” uniforms versus “away” uniforms.

[0049] The same problems are encountered with uniforms worn in commercial endeavors, such as car dealerships, body shops, delivery services, etc.-any business where a customer-facing employee wears a uniform that includes both fixed and custom elements: for example, uniforms that include a fixed company name or logo on a patch or embroidered on the uniform, and a custom name tag or designation for that employee.

[0050] The method disclosed herein also enables printing short runs of uniforms for special occasions or events. As shown in the figures, the variable image printed into the printable field **16** is a number with a star decorative element. That's solely for exemplary purposes and is non-limiting. The image can literally be of anything that can be printed via sublimation printing. Thus, short runs of articles that commemorate or acknowledge some event or person or place, etc., can be fabricated quickly, with very high-quality results, and in small runs—while still being profitable to the manufacturer.

[0051] A problem encountered when heat transferring an image to a garment is that the edges of the imprinted image do not blend well with the color of the shirt itself. In this situation, the resulting garment is amateurish in appearance. It looks as one would expect it to look—a shirt with an off-color image haphazardly affixed to it. In the present method, the image to be affixed to the shirt is applied to one of the light-colored printable fields **16**. In FIG. **1B**, the blending problem is solved by closing matching color or pattern **18** on the garment itself with the overlapping portion **18'** of the image to be transferred. By making the image on the transfer sheet **19** very slightly larger in area than the printable field **16** into which it will be printed, and including in the image to be transferred the color-matched portion **18'** (which matches the fixed background **18** on the garment), the border between the applied image and the remainder of the garment is imperceptible. The final garment, as shown in the lower-right panel of FIG. **1B**, has a high-quality, professional appearance.

[0052] The printing process itself is straightforward. A desired image contained in a computer file is printed (typically using a conventional ink-jet printer using dye sublimation inks) onto a conventional heat transfer sheet. (Heat transfer paper and polymeric heat transfer sheets are widely available from a host of commercial suppliers, such as Coastal Business Supplies, Maryland Heights, Missouri, and many others.) The image is then transferred, in registration, from the transfer sheet **19** to the printable field **16**.

[0053] FIGS. **2A** and **2B** show another version of the method. Here, printable field **16** is surrounded by a gradient **24** that masks the border between the art being applied and the fixed background color or pattern **18** of the garment. As shown in FIG. **2B**, the size of the heat transfer

sheet on which the image appears is slightly larger than the image itself. It too contains an gradient **26** that runs in an opposite direction from the gradient **24** on the shirt itself. In short, to make a smooth transition between the applied image and the remainder of the garment, a pair of cooperating gradients are provided around the outside of the printable field **16** of the article of clothing and also around the outside of the artwork that is to be heat transferred to the article of clothing. These cooperating gradients are shown in FIGS. **2A** and **2B**. FIG. **2A** shows a color gradient (or rather in this situation, a grey-scale gradient) **24**, from dark to light (moving from **18** toward **16**; i.e., outward to inward), circumscribing the printable field **16** of the shirt. That is, the gradient gets lighter moving from outside the printable field **16** toward the inside of the printable field. As shown in the top panel of FIG. **2B**, a corresponding gradient **26**, moving in the other direction, is then provided on the print side **20'** of the image transfer sheet **19**.

[0054] To transfer the image, the gradient **26** on side **20'** of the transfer sheet **19** is placed onto the garment **10** in face-to-face orientation and in registration with both the printable field **16** and the inverse gradient **24** on the article of clothing. See FIG. **2B**. Here, artwork to be applied has a corresponding, but inverse color gradient (or grey-scale gradient) around the perimeter of the artwork. When the artwork is applied to the garment, the two gradients are placed in registration and “offset” one another. The result is a smooth, essentially imperceptible border region between where the artwork has been applied and the remainder of the garment. See FIG. **2B**, lower-right panel, which shows the finished product after the artwork has been applied. The border between the fixed background **18** of the shirt and the printed area is imperceptible due to the oppositely oriented gradients on the shirt and the applied artwork. As in the first version of the method, the image to be transferred includes a surrounding portion **18'** that matches the fixed color or pattern **18** on the garment being printed. See the top panel of FIG. **2B**.

[0055] Note also in FIG. **2B**, lower-left panel, it is not required that a reverse gradient appear on the front side **20'** of the transfer sheet. While a reverse gradient **26** may appear on the transfer sheet in this version of the method, it is not required. Depending upon the color combinations being printed, the gradient **24** on the shirt is sometimes sufficient to yield a final product in which there is no visually perceptible transition between the custom-printed image **22** and the fixed background **18**.

[0056] The gradients on either the garment or the transfer sheet need not be “linear” or “regular.” That is, they need not fade from light to dark (or dark to light) in any regular or predictable fashion. They may, for example, be irregular in appearance, even though they embody an overall gradient of hue or contrast from dark to light or vice-versa.

[0057] A third version of the method is illustrated in FIGS. **3A** and **3B**. Again, the same reference numerals are used in these two drawings. But here, FIG. **3A** depicts a shirt and corresponding method of printing artwork onto the shirt wherein the printable field(s) **16** are surrounded by or circumscribed by a decorative border **30** that masks the transition zone from the custom-applied artwork and the fixed background color or pattern **18** of the shirt. The decorative border **30** may be defined in either the shirt itself (surrounding the printable field **16**) (see FIG. **3A**, right-hand panel) or in the transfer sheet (see FIG. **3B**, top panel), or both. This gives a “picture frame”-type appearance to the applied artwork. See FIG. **3B**, lower-right panel.

[0058] FIGS. **4A**, **4B**, and **4C** illustrate the same method as disclosed above in FIGS. **1A** and **1B**, but here using a hat **40**, rather than a shirt **10**. In the same fashion, the hat **40** includes at least one printable field permanently situated within a fixed background or pattern **18**. As shown in FIG. **4B**, top panel, a transfer sheet **19** includes both a back side **20** and a printed side **20'** having printed thereon an image using ink that can be transferred by sublimation. Surrounding the image is a color matched area **18'** that matches the color **18** in the fixed background portion of the hat. As described above, the printed side **20'** of the transfer sheet **19** is contacted in registration with the printable field **16** in face-to-face orientation, slightly overlapping the edges. The image on the transfer sheet is then transferred to the printable field **16** by the application of heat and pressure for a time to

sufficient to transfer the image. The finished product is shown in FIG. 4C. As shown in the figure, there is no discernable transition between the printed image **42** and the fixed background color **18**. [0059] In all of the versions of the method described, the sublimation printing process has been accomplished using a transfer sheet. That is, the image to be printed is first printed onto a transfer sheet and then transferred (using a heat press) onto the article of clothing. This is for illustration purposes only. The image may also be transferred to the printable field **16** by direct-to-fabric sublimation printing, without a transfer sheet. The direct printing method proceeds in the same fashion as described hereinabove. But rather than printing the image to a transfer sheet, the printable field **16** is placed into a sublimation printer in registration with (or suitably oriented to) one or more printheads of the printer so that the image can be directly printed onto the clothing article (and in the absence of a transfer sheet).

[0060] In all versions of the method, the various panels of the article of clothing (i.e., for a shirt, the front panel, back panel, sleeves, placket, etc.) can be the same color or pattern or of different colors or patterns.

Claims

1. A method of affixing indicia or artwork to an article of clothing, the method comprising: providing an article of clothing comprising at least one, permanently defined or attached printable field having a shape, a size, and a perimeter, wherein the printable field is lighter in color than a surrounding area of the article of clothing, and wherein the surrounding area of the article of clothing is a fixed, pre-fabricated color or pattern; providing a heat-transferable image having a shape, a size, and a perimeter, wherein: the shape of the heat-transferable image corresponds to the shape of the printable field; the size of the heat-transferable image is the same as or larger than the size of the printable field; and the perimeter of the heat-transferable image is a color or pattern substantially identical to the fixed, pre-fabricated color or pattern of the surrounding area of the article of clothing; and heat transferring the heat-transferable image in registration onto the printable field to yield a heat-transferred image having a perimeter, wherein there is no visually discernible border between the perimeter of the heat-transferred image and the perimeter of the printable field.
2. The method of claim 1, comprising providing the heat-transferable image on a heat transfer sheet.
3. The method of claim 1, wherein the article of clothing comprises polyester.
4. The method of claim 3, wherein the article of clothing is a shirt.
5. The method of claim 3, wherein the article of clothing is a hat.
6. The method of claim 1, wherein the printable field is white.
7. The method of claim 1, wherein the printable field comprises undyed polyester.
8. A method of affixing indicia or artwork to an article of clothing, the method comprising: providing an article of clothing comprising at least one, permanently defined or attached printable field having a shape, a size, and a perimeter, wherein the printable field is lighter in color than a surrounding area of the article of clothing, and wherein the surrounding area of the article of clothing is a fixed, pre-fabricated color or pattern, and wherein the perimeter of the printed field comprises a color or grey-scale gradient; providing a heat-transferable image having a shape, a size, and a perimeter, wherein: the shape of the heat-transferable image corresponds to the shape of the printable field; the size of the heat-transferable image is the same as or larger than the size of the printable field; and the perimeter of the heat-transferable image comprises a color or grey-scale gradient inverse to and in registration with the color or grey-scale gradient on the perimeter of the printable field of the article of clothing; and heat transferring the heat-transferable image in registration onto the printable field to yield a heat-transferred image having a perimeter, wherein there is no visually discernible border between the perimeter of the heat-transferred image and the

perimeter of the printable field.

9. The method of claim 8, comprising providing the heat-transferable image on a heat transfer sheet.

10. The method of claim 9, wherein the article of clothing is a shirt.

11. The method of claim 9, wherein the article of clothing is a hat.

12. The method of claim 8, wherein the printable field is white.

13. The method of claim 8, wherein the printable field comprises undyed polyester.

14. A method of affixing indicia or artwork to an article of clothing, the method comprising: providing an article of clothing comprising at least one, permanently defined or attached printable field having a shape, a size, and a perimeter, wherein the printable field is lighter in color than a surrounding area of the article of clothing, and wherein the surrounding area of the article of clothing is a fixed, pre-fabricated color or pattern, and wherein the perimeter of the printed field comprises a decorative motif; providing a heat-transferable image having a shape, a size, and a perimeter, wherein: the shape of the heat-transferable image corresponds to the shape of the printable field; the size of the heat-transferable image is the same as or larger than the size of the printable field; and the perimeter of the heat-transferable image comprises a decorative motif corresponding to and in registration with the decorative motif on the perimeter of the printable field of the article of clothing; and heat transferring the heat-transferable image in registration onto the printable field to yield a heat-transferred image having a perimeter, wherein the decorative motif on the perimeter of the heat-transferable image is transferred in registration onto the decorative motif on the perimeter of the printed field.

15. The method of claim 14, comprising providing the heat-transferable image on a heat transfer sheet.

16. The method of claim 15, wherein the article of clothing is a shirt.

17. The method of claim 15, wherein the article of clothing is a hat.

18. The method of claim 14, wherein the printable field is white.

19. The method of claim 14, wherein the printable field comprises undyed polyester.

20. An article of clothing produced by the method of claim 1.
